

Homework 01

ENME 485
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Industry 4.0

Industry 4.0 is the fourth industrial revolution, and its central idea is adding intelligence to industrial systems, such that they can constantly be improving.

- Industry 1.0
 - Application of steam power
 - Mechanization
- Industry 2.0
 - Use of electricity
 - Mass production
- Industry 3.0
 - Use of computers
 - Automation
- Industry 4.0
 - Add intelligence to industry
 - Use IoT, CPS, AI/ML, etc.

Industry 3.0 and the use of the computer laid out the technological framework for the use of information technology in industry.

Industry 4.0 takes collected information and turns it into intelligence, allowing for self-aware and constantly improving systems to be made.

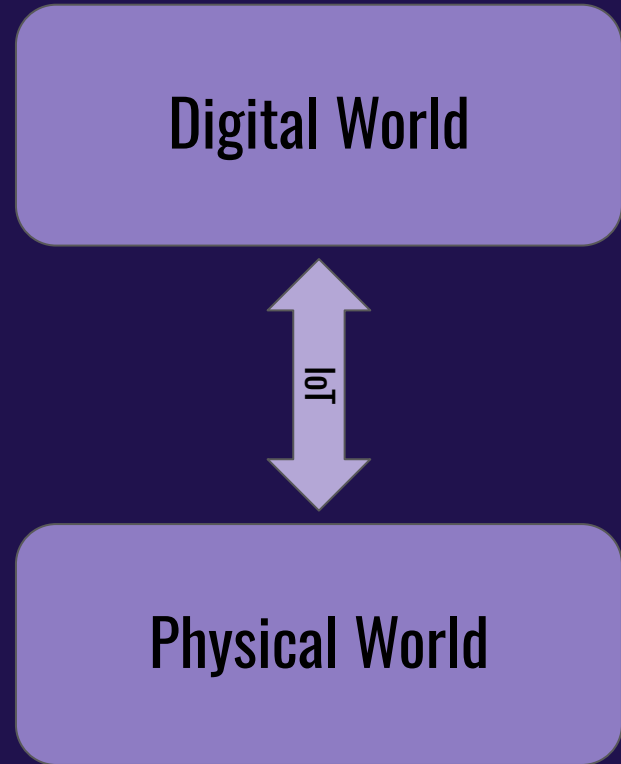
Internet-of-Things (IoT)

The Internet-of-Things is a concept of a network of devices that connect the physical world to the digital world.

An IoT device must have:

- At least one sensor/actuator
- A network interface

Notably, a closed feedback loop is not necessary to make an IoT device. A sensor that just reads and uploads data still qualifies.

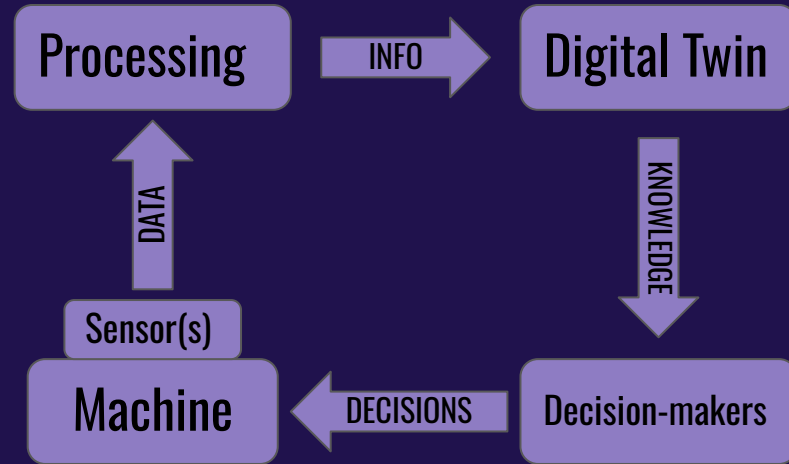


Cyber-physical Systems (CPS)

A Cyber-physical system is an engineered system of a similar nature to an IoT device, but with the added depth of a full digital representation of the system's performance, as well as a closed feedback loop allowing reported data to be used to improve the configuration of the system.

5C Architecture:

- Connect
- Convert
- Cyber
- Cognition
- Configuration



Smart Manufacturing

Smart Manufacturing a manufacturing paradigm in which technologies like IoT and CPS are integrated into factories in order to allow them to create better models of themselves, have a better understanding of the performance of individual machines, as contextualized by data from the entire fleet, and ultimately to make factories more self-aware and adaptable.

Individual machines:

- Collect data
- Compare performance to other machines
- Predict failures

Manufacturing system:

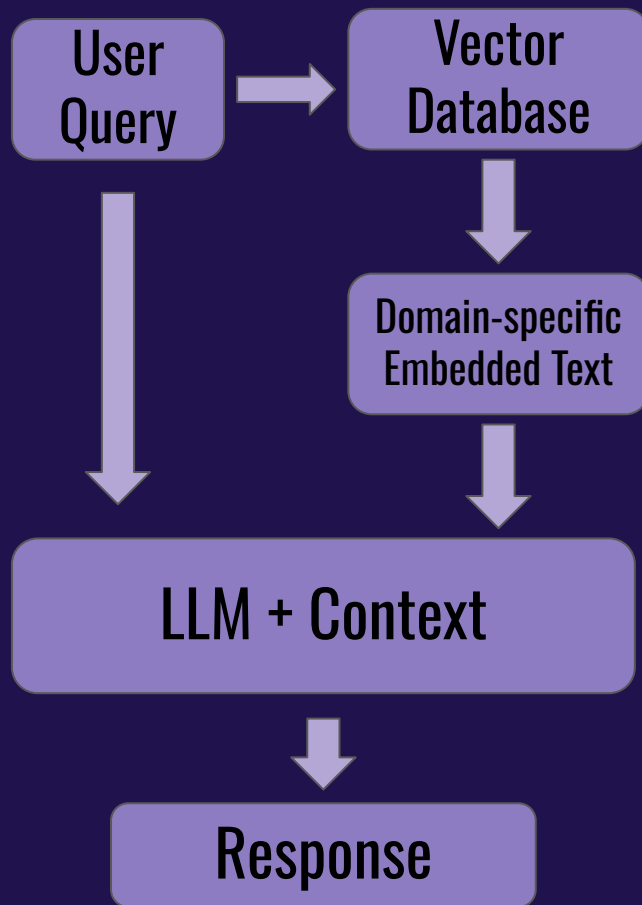
- Interpret data from all machines
- Schedule maintenance and minimize downtime
- Identify areas for change in machine settings to improve overall performance

Large Language Models (LLMs)

LLMs are statistical tools trained to find patterns in human language. Typically using the Transformer neural network architecture, they predict likely tokens based on their input and billions of parameters, and then report output.

LLMs can struggle with answering domain-specific questions due to a lack of training in that domain. Techniques such as Retrieval-Augmented Generation (RAG) can help to provide this context, and allow a general-use LLM to answer domain-specific questions.

RAG Architecture:



Industrial Big Data

Industrial Big Data is the concept of recording the datasets produced in industry, and transforming them into useful information for decision-making.

Industrial Big Data is distinct from traditional Big Data in that the data it works with is centered around machine and process-generated data, rather than human-generated data.

Industrial Big Data is enabled by technologies such as IoT and CPS, and in turn enables AI/ML tools, and the overall information flows and development frameworks inherent to Industry 4.0.

Typically data volume is naturally large in industrial environments, so creating high quality data becomes the primary challenge with Industrial Big Data.

END

(see report for sources)